31 March 1972

Superseding
MIL-M-55670(EL)
1 May 1969

MILITARY SPECIFICATION

MAST AB-621( )/G

1. SCOPE

1.1 This specification covers the requirements for a quick erectable, 50 foot, uniform sectioned, guyed mast. The mast consists of a launching device for upright erection, eight uniform type mast sections with all necessary hardware, and provision for storage and transit packaging. (See 6.1 & 6.4)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	Steel Strapping, Flat
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Boxes, Wood, Nailed and *Lock-Corner*
PPP-B-636	Box, Fiberboard
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper Water Resistance (for Carton Sealing)

FSC 5985

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MIL-P-116	Preservation; Methods of
MIL-W-6858	Welding, Resistance; Aluminum, Magnesium, Nonhardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys, and Titanium Alloys; Spot and Seam
MIL-W-8604	Welding of Aluminum Alloys: Process for
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-C-13924	Coating, Oxide, Black, for Ferrous Metals
MIL-F-14072	Finishes for Ground Signal Equipment

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-1261	Welding Procedures for Constructional Steels

DRAWINGS

ELECTRONICS COMMAND

SM-B-561775	Tensioner, Guy Wire
SC-DL-561799	Drawing and Data List for Mast AB-621( )/G

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(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS

3.1 First Article.- When specified in the contract or purchase order, the contractor shall furnish First Article units in accordance with 4.3.1.

3.2 Construction.- The mast shall be constructed in accordance with the requirements of this specification and the drawings listed on SC-DL-561799. (See 3.20 & 4.24)

3.3 Cleaning.- (See 3.20)

3.3.1 Parts.- After fabrication, parts shall be cleaned in accordance with requirement 9 of MIL-STD-454 except as specified in an applicable document.

3.3.2 Units.- After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, such cleaning shall also be performed before final assembly of the equipment.

3.4 Finish.- The mast shall be finished in accordance with MIL-F-14072 and the drawings listed on SC-DL-561799. Exterior stainless steel shall be black oxide coated conforming to MIL-C-13924. (See 3.20 & 4.4) The final paint film shall be olive drab semigloss enamel matching a color chip provided by the procuring agency. (See 6.6)

3.5 Marking.- Marking shall conform to MIL-M-13231 and the drawings listed on SC-DL-561799. (See 3.20 & 4.4)

3.6 Riveting.- Riveted joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces. (See 3.20)

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3.7 Securing of parts.- Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall be suitably tight to retain items securely when subjected to specified tests. (See 3.20)

3.8 Tropicalization.-

3.8.1 Treatment of Materials.- Fungus-inert materials shall be used where practicable. These materials shall be in accordance with requirement 4 of MIL-STD-454. When materials are required to be treated, they shall be cleaned in accordance with requirement 9 of MIL-STD-454 and then shall be treated in accordance with 3.8.2 thru 3.8.5 and as follows:

Linen thread, cotton thread and duck (See 3.17 of MIL-P-11268)
Cork (See 3.14 of MIL-P-11268)
Felt, hair or wool (See 3.18 of MIL-P-11268)
Leather (See 3.20 of MIL-P-11268)
Paper, paperboard, cardboard, organic fiberboard, vulcanized fiber, etc. (See 3.19 of MIL-P-11268)
Thermosetting plastic materials using cotton, linen, or wood-flour as a filler or base (See 3.23 of MIL-P-11268)

3.8.2 Treating compounds.- Treating compounds containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating compound is compatible with the material or surface to be treated. Selection of treating compound shall be such that there shall be no increase in flammability of the treated material.

3.8.3 Toxicity.- Treating compounds shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.8.4 Flexibility.- Treatment shall not affect the flexibility of the treated material, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

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3.8.5 Statement of treatment.- The statement of treatment shall indicate whether or not nutrient materials have been employed. If nutrient materials have been used, it shall also describe in detail the materials to be treated and the treating compounds and processes that the contractor proposes to use. (See 6.2i)

3.8.6 Fungus.- The mast shall show no evidence of viable fungus or corrosion when tested per 4.19. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

3.9 Welding.- Welding shall conform to the drawings listed on SC-DL-561799 and to: MIL-W-8604 for inert-gas-shielded arc welding of aluminum alloy, MIL-STD-1261 for metal-arc welding of steel, and MIL-W-6858 for spot, stitch and seam welding. (See 3.20 & 4.4)

3.9.1 Cleaning prior to welding.- Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material.

3.9.2 Process.- Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects.

3.9.3 Cleaning after welding.- Welded assemblies shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Acid used for cleaning shall be completely neutralized and removed.

3.10 Winch.- The winch shall be capable of withstanding the following load conditions:

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3.10.1 Static condition.- The winch shall be capable of supporting a 1,000 pound vertical load on the elevator without movement of the crank handle or permanent deformation of any part of the equipment. (See 4.7.1)

3.10.2 Dynamic condition.- With the elevator at its midpoint of travel, supporting a 370 pound vertical load, the winch shall be capable of raising the elevator with a handle force not to exceed 28 pounds. (See 4.7.2)

3.11 Brake.- The brake, when locked on any portion of the mast section, shall be capable of withstanding an axial force of 850 pounds without yield or slippage. The force required to apply the brake shall not exceed 30 pounds. (See 4.8)

3.12 Indexing (mast sections).- The torsional transfer elements at the mast section ends shall be so fitted that the total angular displacement over the total length of any two coupled mast sections shall not exceed ±10 minutes when subjected to a 300 inch pound torque. (See 4.11)

3.13 Bending (mast sections).- The joint between any two coupled mast sections shall sustain a bending moment of 75,000 inch pounds with no yield or failure of any part of the mast sections. (See 4.12)

3.14 Tensioners.- Each guy shall contain a tensioner per drawing SM-B-561775. The tensioners shall be tested per 4.10.

3.15 Service conditions.- The mast shall sustain no corrosion, winch lubricant leakage, pitting, blistering or loosening of finish or metal surfaces, hardening, cracking, softening or tackiness of rubber or plastic materials, decomposition of cloth, bonding failure, binding of moving parts, deformation, or other deterioration of materials or operation when subjected to the following service conditions:

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3.15.1 Temperature.- Exposure to: (See 4.20)

a. Operating. Ambient temperature in the range of +125° to -25°F. Exposure at both extremes not to exceed 4 hours at any one time.

b. Non-operating. Temperature in the range of +160°F to -65°F. Exposure at both extremes for a minimum of 24 hours.

3.15.2 Humidity.- Exposure to warm, highly humid (tropical) atmosphere. (See 4.21)

3.15.3 Salt atmosphere.- Exposure to salt (coastal) atmosphere. (See 4.22)

3.15.4 Dust atmosphere.- Exposure to sand (desert) or dust atmosphere. (See 4.18)

3.15.5 Altitude.- Exposure to reduced pressure (air shipment or high ground operation). (See 4.17)

3.16 Bounce and shock.-

3.16.1 Bounce, loose cargo.- The mast shall sustain no physical damage except minor surface abrasion when tested per 4.14. Degradation in performance is not permissible.

3.16.2 Shock, drop.- The mast shall sustain no structural damage when tested per 4.15. Denting of the point of contact is permissible. Weld cracks, connector failures, deformation of any member, or degradation in performance is not permissible.

3.17 Mechanical binding.- The mechanical elements of the mast shall be free from binding and chafing as follows:

3.17.1 Operation.- The brake, bearings, winch with mounting system, elevator assembly, azimuth, carrier assemblies, and tensioners shall operate smoothly at specified service conditions. (See 4.23)

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3.17.2 Rotation.- Frictional resistance in the guy rings and bearings shall be such that the mast, with antenna installed, can be rotated while exposed to a 50 mph wind velocity. The mast shall be capable of 360° rotation by use of the spanner tool and the required torque shall not exceed 70 foot pounds. (See 4.13)

3.18 Static loading.- The mast, with antenna installed, shall be capable of withstanding the following loading conditions:

3.18.1 Fully guyed.- The mast, when fully guyed, shall withstand a 100 mph wind velocity with no failure or permanent deformation to any part. (See 4.16.1)

3.18.2 Launcher guyed.- The mast, when guyed at the launcher only, shall withstand a 30 mph wind velocity with no failure or permanent deformation to any part. (See 4.16.2)

3.19 Interchangeability.- All like parts shall be physically and functionally interchangeable, without modification of the parts. Individual parts shall not be hand picked for fit performance. Reliance shall not be placed on any unspecified dimension, rating, characteristics, etc. (See 4.9)

3.20 Workmanship.- The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and with the following paragraphs herein: (See 4.6)

3.2 Construction
3.3 Cleaning
3.4 Finish
3.5 Marking
3.6 Riveting
3.7 Securing of parts
3.9 Welding

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- The inspection requirements specified herein are classified as follows:

a. First Article inspection (does not include preparation for delivery). (See 4.3)

b. Inspections covered by subsidiary documents. (See 4.4)

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.25)

4.3 First Article.- Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor.

4.3.1 First Article units.- The contractor shall furnish three (3) First Article units of the complete Mast AB-621 ( )/G.

4.3.2 First Article inspection.- The first article inspection shall consist of inspections specified in Table I, and shall be performed in the order specified therein.

4.3.3 First Article data.- The First Article test plan and test report shall be as required in the contract.

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TABLE I - FIRST ARTICLE INSPECTION

INSPECTION	REQ'T PARA	INSP PARA	Order of Testing for Unit		
			#1	#2	#3
Subsidiary documents	-	4.4	1	1	1
Visual and mechanical	3.20	4.6	2	2	2
Winch	3.10	4.7	3,13	3	3
Brake	3.11	4.8	4	4	4
Interchangeability	3.19	4.9	5	5	5
Tensioner	3.14	4.10	*	6	6
Indexing	3.12	4.11	6	*	*
Bending	3.13	4.12	7	*	*
Rotational resistance	3.17.2	4.13	8	*	*
Bounce,loose cargo	3.16.1	4.14	9	*	*
Shock, drop	3.16.2	4.15	10	*	*
Static loading	3.18	4.16	11	*	*
Altitude	3.15.5	4.17	12	*	*
Dust	3.15.4	4.18	14	*	*
Fungus	3.8.6	4.19	15	*	*
Temperature	3.15.1	4.20	*	7	*
Humidity	3.15.2	4.21	*	8	*
Salt fog	3.15.3	4.22	*	*	7

*Omit

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4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection during production and before preparation for delivery:

3.4 Finish
3.5 Marking
3.9 Welding

4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 thru 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government, at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.7) Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection.- Each unit produced shall be inspected for conformance to all the inspection requirements of Table II. Each lot shall then be subject to verification, utilizing the procedures of MIL-STD-105 using the general inspection levels and the AQL's indicated in Table II. Group A inspection shall be performed in any order which is satisfactory to the Government, except that visual and mechanical inspection shall be first.

TABLE II - GROUP A INSPECTION

INSPECTION	REQ'T PARA	INSP PARA	AQL	
			MAJOR	MINOR
Visual and Mechanical	3.20	4.6	1%	4%
Winch	3.10	4.7	1%	--
Brake	3.11	4.8	1%	--

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4.5.2 Group B inspection.- This inspection, including sampling, shall conform to Table III and to the special procedures for small-sample inspection of MIL-STD-105. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met Group A inspection. Inspection level shall be S-4 for normal inspection and S-3 for reduced inspection.

TABLE III - GROUP B INSPECTION

INSPECTION	REQ'T PARA	INSP PARA	AQL
Interchangeability	3.19	4.9	6.5%

4.5.3 Group C inspection.- This inspection shall be as listed in Table IV and shall normally be performed on sample units that have been subjected to and met Group A and Group B inspection. Sample units shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspection.- Sample units shall be selected at random for each Group C inspection listed in Table IV in accordance with the following:

4.5.3.1.1 Subgroup I.- One unit from the first 40 production units or fraction thereof and one unit from each successive 40 production units or fraction thereof shall be subjected to the test of subgroup I.

4.5.3.1.2 Subgroup II.- One unit from the first 200 production units or fraction thereof and one unit from each successive 200 or fraction thereof shall be subjected to the tests of subgroup II in the order given in Table IV.

4.5.3.1.3 Subgroup III.- Three units from the first 40 production units or fraction thereof shall be subjected to the tests of subgroup III as follows:

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a. One unit shall be subjected to the altitude, winch, dust and fungus tests in that order.

b. A second unit shall be subjected to the temperature and humidity tests in that order.

c. A third unit shall be subjected to the salt fog test.

4.5.3.1.4 Subgroup IV.- The EVR, per 4.24, shall be performed once during the production of the first lot of 40 units. In the event that discrepancies are found, the contractor shall: (1) immediately halt production of the type of part or assembly on which the discrepancies were found; (2) immediately notify the Government technical activity designated in the contract and include proposed corrective action to eliminate the defective material, parts, subassemblies, assemblies, tooling, process, or procedures; (3) implement corrective measures to assure conforming product. Re-inspection and requalification of the discrepant area(s) is required before proceeding with production.

4.5.3.2 Noncompliance.- The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on the process and materials, as applicable, and all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the quality assurance representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections or the inspections which the sample failed, at the option of the Government). However, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

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TABLE IV - GROUP C INSPECTION

Inspection	REQ'T PARA	INSP PARA
Subgroup I		
Tensioner	3.14	4.10
Subgroup II		
Indexing	3.12	4.11
Bending	3.13	4.12
Rotational resistance	3.17.2	4.13
Bounce, loose cargo	3.16.1	4.14
Shock, drop	3.16.2	4.15
Static loading	3.18	4.16
Subgroup III		
Altitude	3.15.5	4.17
Dust	3.15.4	4.18
Fungus	3.8.6	4.19
Temperature	3.15.1	4.20
Humidity	3.15.2	4.21
Salt Fog	3.15.3	4.22
Subgroup IV		
Equipment Verification Review (EVR)	3.2	4.24

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4.5.4 Reinspection of conforming Group B and Group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed Group B or Group C inspection may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all visible damage.

4.6 Visual and mechanical inspection.- The mast, with accessories, shall be examined for the requirements of workmanship per 3.20 and the defects listed in Table V.

TABLE V
CLASSIFICATION OF VISUAL AND MECHANICAL DEFECTS

Classification		Defect
Major	1.	Welds: cracks, porosity, undercuts, gaps, burn thru or distortion.
	2.	Rivets: loose or heads not fully formed.
	3.	Threaded fasteners: loose
	4.	Threads: cross threaded, burred or mutilated.
	5.	Locking screw and locking pins inoperable.
	6.	Bags: buckles and snaps improperly fitted.
	7.	Azimuth ring or pointers not properly installed.
	8.	Winch mounting clips inoperable.
	9.	Launcher rollers not freely rotating.
	10.	Indexing marks and holes improperly located.
	11.	Indexing pin inoperable.
	12.	Upper guy ring not freely rotating.
	13.	Dry film lubricant missing from mast cap, middle guy ring or mast section grooves.
	14.	Guy cable jacketing cracked.
	15.	Rubber bumpers cracked or improperly attached.
	16.	Nameplate, installation or brake caution plates incorrect or illegible.
Minor	1.	Finish: abrasions, scratches or flakiness.
	2.	Guys not properly color coded.
	3.	Cloth labels incorrect or illegible.
	4.	Equipment not free of foreign material, rough finishes.

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4.7 Winch tests.- The mast shall be subjected to the following tests:

4.7.1 Static test.- With the launcher in the vertical position, a mast section in the launcher engaging the elevator, the brake disengaged, and the elevator at its midpoint of travel, a 1,000 pound axial load shall be applied to the mast section. With no external restraint, the winch shall meet the requirements of 3.10.1.

4.7.2 Dynamic test.- With the launcher in the vertical position, a mast section in the launcher engaging the elevator, the brake disengaged, and the elevator at its midpoint of travel, a 370 pound axial load shall be applied to the mast section. The elevator shall be raised thru a distance of one foot by applying a force to the winch handle. The force shall be measured throughout the travel and shall at no time exceed the requirement of 3.10.2.

4.8 Brake test.- With the launcher in the vertical position and a mast section in the launcher, the force required to engage the brake shall be measured and shall not exceed the requirement of 3.11. This force shall be applied at the lock pin hole in the handle. The elevator shall then be disengaged and an 850 pound axial load applied to the mast section. The requirements of 3.11 shall be met.

4.9 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.19. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. Dimensions and location of torsional transfer elements.

b. Mating dimensions of section end fittings.

c. Mounting dimensions of separable assemblies.

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4.10 Tensioner tests.- Three tensioners shall be randomly selected from each mast sample and each tensioner shall be subjected to the following tests: (See 3.14)

4.10.1 Safety lock.- The clamp handle shall be moved to the closed position and the safety lock shall engage the handle with no external aid. With no further setting of the safety lock, an opening force of 10 pounds shall be applied to the free end of the clamp handle. The safety lock shall keep the handle from opening.

4.10.2 Clamp handle.- With guy cable engaged (no-tension) and safety lock disengaged (locked open), the clamp handle shall be operated from fully open to fully closed by applying a force at its free end. The required force shall be measured and shall not exceed 31 pounds.

4.10.3 Tensioning torque.- With 300 pounds tension applied to the tensioner thru the guy cables, the turnbuckle nut shall be turned in the direction of increasing cable tension. The required torque on the turnbuckle nut shall be measured and shall not exceed 140 inch pounds.

4.10.4 Slipping strength.- With the safety lock disengaged, 2,000 pounds tension shall be applied to the tensioner thru the guy cables. There shall be no yielding of the tensioner or slippage of the cable in the tensioner.

4.10.5 Breaking strength.- With the safety lock disengaged, 3,000 pounds tension shall be applied to the tensioner thru 3/16 inch diameter bare steel cable. There shall be no yielding of the tensioner or slippage of the cable in the tensioner.

4.11 Indexing test.- A mast section shall be fixed at one end and freely supported at the other. A second mast section shall be coupled to the freely supported end and freely supported at its other end. At this end a 305 inch pound torque shall be applied both in a clockwise and counterclockwise direction. The total angular displacement (including tolerance takeup and torsional strain) between these two extremes shall be measured and shall not

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exceed 20 minutes. (See 3.12) The fixed end shall then be freed and the torque required to rotate the masts in the two supports shall be measured. This torque shall not exceed 10 inch pounds.

4.12 Bending test.- Two mast sections shall be coupled together and simply supported with a span measuring 120 inches inside to inside of supports. A 2,500 pound load shall be applied at the center of the 7 inch joint overlap, which shall be positioned centrally in the span. The supports and load shall each contact the mast sections for no more than 120° of circumference and 3 inches of length. The keys and keyways of the joint shall be in the same plane as the applied load. The mast sections shall meet the requirements of 3.13.

4.13 Rotational resistance test.- The mast, completely extended (brake disengaged), fully guyed and rigged for operational use and carrying a 190 pound load located at the mast top, 12 inches from the center of the mast, shall be statically loaded by applying tension to the guys as follows:

a. 250 pounds to each of four upper guys

b. 120 pounds to each of four mid guys

c. 280 pounds to each of four lower guys

The mast shall be rotated through 360° by applying a torque using the spanner tool. The torque shall be measured throughout the travel and shall at no time exceed the requirements of 3.17.2.

4.14 Bounce, loose cargo test.- The launcher, with mast sections and carrier prepared for transit, and the stake carrier, with stakes and hammer prepared for transit, shall be tested per Method 514, Procedure XI, Part 2 of MIL-STD-810 as follows:

a. The total horizontal motion (both directions) shall be limited to not more than 2 inches and not less than 1 inch by wooden fences at least as high as the equipment.

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b. The package tester shall be operated in the vertical linear mode only.

c. The launcher and stake carrier shall be tested separately or if tested simultaneously they should be fenced separately.

d. The launcher, with mast sections and carrier, shall be bounced for 45 minutes in each of 4 positions. The positions shall be: while resting on each of 3 sides (with direction of elevator travel horizontal, but not including the side with the mast carrier handles) and while resting on the base.

e. The stake carrier, with stakes and hammer, shall be bounced for 90 minutes in each of two positions. The positions shall be: while resting on each of 2 sides (with plane of stakes horizontal).

After test, the mast shall meet the requirements of 3.16.1.

4.15 Shock, drop test.- The launcher, with mast sections and carrier prepared for transit, shall be tested per Method 516, Procedure II of MIL-STD-810 as follows:

a. The weight of the test item shall be taken as over 200 pounds and less than 1,000 pounds. Its largest dimension shall be taken as over 60 inches.

b. Two drops on each corner, of the side on which the winch mounts, shall be performed for a total of 8 drops. After test, the mast shall meet the requirements of 3.16.2.

4.16 Static loading tests.- The mast shall be subjected to the following tests.

4.16.1 Mast fully guyed.- The mast, completely extended (brake engaged), fully guyed and pretensioned with 150 pounds in each guy, shall be simultaneously loaded with the following forces:

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a. 190 pound weight applied at the mast top 12 inches from the center of the mast.

b. 1,100 pound horizontal pull applied at a distance of 42 inches above the top of the mast.

c. Horizontal pulls of 70 pounds each applied at the midpoint of each mast section.

All of the forces shall be applied on the same side of the mast in a plane which bisects the angle between two adjacent guys. After test, the mast shall meet the requirements of 3.18.1.

4.16.2 Mast guyed at launcher only.- The mast, completely extended (brake engaged), with the four lower level guys pretensioned with 150 pounds each and with all other guys removed, shall be simultaneously loaded with the following forces:

a. Same as in 4.16.1a.

b. 58 pound pull applied as in 4.16.1b.

c. 5.5 pound pulls applied as in 4.16.1c.

All of the forces shall be applied on the same side of the mast and in the same plane. After test, the mast shall meet the requirements of 3.18.2.

4.17 Altitude test.- The winch shall be tested per Method 500, Procedure I of MIL-STD-810. The winch handle shall be cranked thru 10 revolutions at Step 3. There shall be no lubricant leakage from any part of the winch during or after test. (See 3.15.5) After test, the mast shall be tested per 4.7 and meet the requirements of 3.10.

4.18 Dust test.- The following mast components shall be tested per Method 510, Procedure I of MIL-STD-810: launcher, mast carrier, 3 mast sections, mast cap, winch, 4 guy assemblies, and both guy rings. The operational inspection of 4.23 (except 4.23i) shall be applicable at Step 5. The mast shall meet the requirements of 3.15. (See 3.15.4)

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4.19 Fungus Test.- The mast, with all accessories, shall be tested per Method 508, Procedure I of MIL-STD-810. Those items normally stored in bags shall be removed and exposed during test. Operation is not required. The mast shall meet the requirements of 3.8.6.

4.20 Temperature tests.- The mast, with all accessories, shall be subjected to the following tests: (See 3.15.1)

4.20.1 High temperature.- Method 501, Procedure I of MIL-STD-810. Step 3 shall be maintained for a period of not less than 24 hours. Temperature for Step 4 shall be +125°F. At Steps 5 and 7, the operational inspection of 4.23 shall be applicable. The mast shall meet the requirements of 3.15.

4.20.2 Low temperature.- Method 502, Procedure I of MIL-STD-810. Temperature for Step 2 shall be -65°F and for Step 4 shall be -25°F. Step 2 shall be maintained for a period of not less than 24 hours. At Steps 5 and 7, the operational inspection of 4.23 shall be applicable. The mast shall meet the requirements of 3.15.

4.21 Humidity test.- The mast, with all accessories, shall be tested per Method 507, Procedure II of MIL-STD-810. The operational inspection of 4.23 shall be applicable at Steps 4 and 8 and during the 24th to 28th hours of each cycle of Step 6. The mast shall meet the requirements of 3.15. (See 3.15.2)

4.22 Salt fog test.- The mast, with all accessories, shall be tested per Method 509 of MIL-STD-810. Those items normally stored in bags shall be removed and exposed during test. The operational inspection of 4.23 shall be applicable after exposure and after drying. The mast shall meet the requirements of 3.15. (See 3.15.3)

4.23 Operational inspection.- The following operations shall be performed, where specified, to determine the results of other tests (See 4.18,4.20,4.21 & 4.22):

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a. Operate lock pins on elevator, brake, and winch reel and locking screw for azimuth ring.

b. Mount winch on launcher.

c. Position (using keyway) one mast section in launcher and operate elevator.

d. Operate brake with mast section in place.

e. Install middle and upper guy rings and mast cap.

f. Rotate azimuth (with mast section) and each guy ring independently.

g. Install 2 mast sections in mast carrier and lock handle.

h. Install mast carrier in launcher and lock in place.

i. Install stakes in stake carrier and lock carrier.

j. On a tensioner operate the turnbuckle nut, the clamp handle and lock, and pull a length of cable thru the tensioner. Any malfunction or deterioration in ease of operation shall be cause for rejection (See 3.17.1) under the test being analyzed.

4.24 Equipment verification review.- The contractor shall perform an equipment verification review (EVR), consisting of a complete technical audit of the equipment on order against the drawings cited for construction. (See 3.2) The EVR shall consist of the following:

a. An audit to establish that "as-built" equipment (unit, assembly, module, or part) is in accordance with the cited end product drawings.

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b. An audit to establish that "as-built" configuration of unit, assembly, module, or part meet the acceptance requirements (other than equipment specification performance) specified for each unit, assembly, module, or part. The audit shall record all acceptance test methods used together with resulting verification test data.

c. The contractor shall compile an EVR report(s) containing the following information:

(1) Identification of unit, assembly, module, or part and details proving compliance with 4.24a and 4.24b above.

(2) Discrepancies noted.

(3) Corrective action taken.

d. The contractor shall make available to the Government copies of his EVR report(s) no later than fifteen (15) days prior to the submission of the first production unit or lot. The results of the EVR shall be subject to inspection by authorized Government personnel at the time the first unit or production lot is offered for acceptance. Government inspection will be to the depth and extent necessary to demonstrate that the "as-build" hardware is in accordance with the cited drawings. Copies of the EVR reports shall be made available to the Government personnel during the inspection. The contractor shall provide the following:

(1) Segregation of units, assemblies, modules, and parts to permit reviewing personnel access for detailed inspection.

(2) As required, responsible personnel from each functional department available for discussions in their respective areas.

(3) Adequate administrative support for the EVR.

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4.25 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

4.26 Rough handling test (preparation for delivery). When the rough handling test in accordance with MIL-P-116 is required by the contract, the operational inspection of ..23 shall be conducted to determine freedom from operational malfunction caused by rough handling.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C as specified. (See 6.2b)

5.1.1 Level A.-

5.1.1.1 Cleaning.- Mast AB-621( )/G shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying.- Mast AB-621( )/G shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application.- None required.

5.1.1.4 Unit packaging.- Unit packaging shall be in accordance with the method prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature.- Each technical literature shall be packaged method 1C-1.

5.1.1.4.2 Mast sections.- Mast sections shall be packaged method III as follows: Place the mast sections within the mast carrier and secure with locking devices provided. Place the loaded mast carrier within the launching frame and secure with fasteners provided. No further packaging required.

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5.1.1.4.3 Stake carrier.- Each stake carrier with stakes and sledge hammer shall be packaged method III as follows: Place the stakes and sledge hammer within the stake carrier and secure locking devices. No further packaging required.

5.1.1.4.4 Accessory bag & stake bag.- Each accessory bag and stake bag with contents shall be packaged method III as follows: Place the stakes within the stake bag and secure bag with straps provided. Place the accessories and the bag containing the stakes within the accessory bag and secure accessory bag with fasteners provided. Place the accessory bag with contents within a close fitting fiberboard box conforming to PPP-B-636, W5c, fill all voids with paperboard conforming to PPP-P-291, type 1, to prevent movement. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.1.4.5 Guy bag.- Each guy bag with contents shall be packaged method III as follows: Place the designated units within the guy bag. Close cover and secure with fasteners provided. Place the guy bag within a close fitting fiberboard box conforming to PPP-B-636, W5c. Closure shall be in accordance with the appendix of the box specification.

5.1.2 Level C.- Mast AB-621( )/G shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited in transit storage without damaging the container or its contents.

5.2 Packing.- Packing shall be level A,B or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container or its contents. (See 6.2b)

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5.2.1 Level A.- Mast sections packaged as specified in 5.1 shall be packed, wood blocked and braced within a close fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. 3x4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of materials handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, class B.

5.2.1.1 Stake carrier, accessory bag, and guy bag.- Each item packaged as specified in 5.1 shall be packed together, wood blocked and braced within a close fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2, or PPP-B-585, style 2 or 3, class 3. Fiberboard boxes being placed directly in the shipping container shall be waterproofed with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. 3x4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, class B.

5.2.2 Level B.- Mast sections packaged as specified in 5.1 shall be packed, woodblocked and braced within a close fitting box conforming to PPP-B-601, domestic type; PPP-B-621, style 4, class 1; or PPP-B-585, style 2 or 3, class 3. 3x4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto.

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5.2.2.1 Stake carrier, accessory bag and guy bag.- Each item packaged as specified in 5.1 shall be packed together, wood blocked and braced within a close fitting box conforming to PPP-B-601, domestic type; PPP-B-621, style 4, class 1; or PPP-B-585, style 2 or 3, class 3. 3x4 inch skids, laid flat shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto.

5.2.3 Level C.- Mast AB-621( )/G packaged as specified in 5.1 shall be packed in shipping containers that comply with Federal or Military specifications, in a manner that will afford adequate protection to the package and its contents against physical and environmental deterioration and damage during limited shipment, handling and in transit storage.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- Mast AB-621( )/G is used for support of Radio Antennas and associated transmission lines.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Levels of preservation and packaging and packing. (See 5)

c. When rough handling and functional tests are required. (See 4.26)

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d. First article inspection:

Three sample units of Mast AB-621( )/G are generally required so that lengthy environmental tests can be completed on two sample units while complete performance measurements can be made on the third sample unit. (See 4.3)

e. Marking and shipping of samples.

f. Place of final inspection.

g. Technical literature required.

h. Quantity of tools and running spare parts required.

i. Submission of the statement of treatment referenced in 3.8.5, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2)

6.4 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AB-621/G or AB-621A/G. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.5 Color of textiles.- Standard samples of olive drab No. 7 may be obtained from the Sample Loan Unit, Defense Clothing and Textile Supply Center, Philadelphia, Pennsylvania 19101. When requesting samples, indication should be given as to type of material concerned.

6.6 Color.- The color chip furnished by the procuring agency will be color chip No. X-24087 and may be obtained upon request to Commanding General, U.S. Army Electronics Command, 225 South Eighteenth Street, Philadelphia, Pennsylvania 19103, Attn: AMSEL-PP/P-CP.

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6.7 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:
Army - EL

Preparing Activity:
Army - EL

Project No. 5985-A374

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MAST AB-621( )/G MIL-M-55670A(EL)

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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